SPE-229857-MS

Success Factors and Design Challenges for Subsea Pipelines Lateral Buckling Mitigation

M. Abdel Hakim, ADNOC, Abu Dhabi, Abu Dhabi, UAE

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Lateral buckling is a key design challenge for High Pressure / High Temperature (HPHT) subsea pipelines. Various mitigation techniques are typically used in the industry. These mitigation techniques are assessed to present the key design challenges associated with each technique and the recommended design practices to overcome these challenges.

Lateral buckling design approaches for subsea pipelines are assessed based on their advantages and limitations. The assessment covers the two key mitigation approaches used in the industry which are buckling prevention and controlled buckling using buckle triggers. The assessment is based on lessons learned and best practices from case studies for subsea pipelines in shallow water applications. The assessment includes general overview of the design methodology, design challenges and limitations for each mitigation approach.

The assessment demonstrates that each lateral buckling mitigation involves specific design challenges. There is no 'one size fits all' design solution to be used in all cases. Hence, the selection of the lateral buckling design solution should be based on comprehensive analysis of the technical feasibility, constructability and cost-effectiveness of the various mitigation techniques. Clear understanding of the design challenges for each mitigation approach and the recommended design practices for overcoming them is a key success factor to achieve a reliable and cost-effective design.

This paper aims to provide guidance for selection of lateral buckling design solution for subsea pipelines based on the merits and shortcomings of the various mitigation techniques.

Introduction

Lateral buckling has been identified as a phenomenon since the 1980's (Hobbs [1]) where high effective axial compressive forces due to build up of soil resistance against thermal expansionleads to buckling of the pipeline as a column in compression i.e. Euler buckling.Lateral bucklingis not a failure mode; however, it can lead to several failure modes which can seriously jeopardize the structural integrity of a subsea pipeline.

The failure modes or limit states associated with lateral buckling include the following (DNF-RP-F110 [2]):

- Local buckling under combined loading i.e. axial compressive force, bending moment and internal overpressure.
- Fatigue and fracture due to crack growth under cyclic tensile stresses.
- Plastic collapse due to longitudinal strain exceedingthe uniform strain capacity corresponding to the material tensile strength.
- Cyclic plasticity due to Bauschinger effect i.e. reduction in yield stress due to cyclic loading.
- Excessive pipeline lateral displacement leading to interference with other subsea infrastructure.
- Overstressing of riser tie-in spool and / or route curve pull-out due to pipeline walking associated with thermal gradients during start-up / shutdown cycles.

Lateral buckling mitigation techniques can generally be classified into the following two main design approaches:

- Buckling prevention by restrainingthe pipeline against lateral buckling.
- Controlled buckling by allowing the pipeline to buckle at predetermined locationsto release the thermal expansion in a "controlled" manner.

Both design approaches involve numerous design challenges and hence overcoming those challenges is instrumental to ensure the effectiveness and reliability of the design solution.

Buckling Prevention

Buckling prevention often representsthe simplest and most straightforwarddesign approach. It is based on ensuring that the critical buckling force is higher than the compressive axial force along the pipeline route. Hence, it is an inherently safer design approach compared to controlled buckling; however, this option is often difficult and expensive particularly for high pressure high temperature pipelines.

Buckling Prevention Techniques

Buckling prevention for subsea pipelines in shallow water applications can typically be achieved through a combination of the following techniques:

- Increase the pipe submerged weight by increasing the concrete weight coating thickness and / or density.
- Reduce the pipe out-of-straightness by increasing the radius of curvature at route bends and decreasing the pipe lay tolerances.
- Restrain the pipeline laterally by installingpre-cast concrete / grout mattresses.

Buckling Prevention Design Challenges

Despite its apparent simplicity, buckling prevention entails significant design challenges due to uncertainty in the following key design parameters affecting the pipeline susceptibility to lateral buckling:

Pipeline Out-of-straightness. Pipeline out-of-straightness includes horizontal out-of-straightness due to route curvature at route bends and pipelay induced imperfections, and vertical out-of-straightness due to seabed irregularity. For new pipelines, route curvatureat bends is known and seabed irregularity can be acquired with high level of certainty through bathymetric survey and accurately included in finite element analysis models. On the other hand, horizontal out-of-straightness along nominally straight sections of the route due to pipelay induced imperfections is far more difficult to predict during the design stage. The horizontal imperfectionsintroduced by routine adjustment of pipelay vessel heading during installation to maintain the pipeline within the specified lay corridorcan only be ascertained after the installation of the pipeline.

In addition, modelling the pipelay induced imperfections in the finite element analysis is less straightforward. Such imperfections can generally be simulated by introducing a lateral "nudge" or a gentle sine curve with amplitude equal to the maximum lay tolerance and wavelength based on the installation contractor experience to realistically mimic the as-laid radius and length of the imperfection. The nudging method is relatively more conservative as it results in high local curvature while the sine wave method entails a significantly high level of uncertainty.

Soil Resistance. Pipe-soil interaction for exposed subsea pipelines is a complexand highly non-linear phenomenon,particularly for calcareous soil prevalent in the Arabian Gulf. The axial and lateral soil resistancesdepend on numerous factors including the soil properties, the pipeline as-laid embedment andthe pipeline loading history. In addition, the type of soil and the soil properties can vary significantly along the pipeline route. Hence, the axial and lateralsoil resistancesalong the pipeline route are typically represented by a wide range defined by low, best and high estimates, which complicates the design process.

Hydrodynamic Forces. Surface laid subsea pipelines in shallow water are subject to substantial hydrodynamic forces due to the action of waves and currents. The drag and lift hydrodynamic forces can significantly increase the pipeline susceptibility to lateral buckling by decreasing the pipe submerged weight and reducing the lateral restraint against buckling. In addition, subsea pipelines in shallow water are typically designed to accommodate limited level of lateral displacement during extreme storm conditions (DNV-RP-F109 [3]). Hence,

Limitations on Additional Weight. For new pipelines, increasing the pipeline submerged weight is limited by a number of factors including the maximum feasible concrete weight coating thickness based on the coating facility capabilities, pipelay vessel tensioner capacity as well as offshore constructabilityto avoid risk of local buckling during laying in case of excessive concrete weight coating thickness.

Buckling Prevention Success Factors

The feasibility and reliability of buckling prevention as a design approach can be significantly improved through adopting the following best engineering practices:

Estimation of Compressive Axial Force. The restrained effective axial force along the pipeline is estimated based on the following equation (DNV-RP-F110 [2]):

$$S_0 = H_{lay} - \Delta p_i \cdot A_i(1 - 2 \cdot \vartheta) - 2 \cdot A_s \cdot E \cdot \alpha \cdot \Delta T$$

where:

H_{lay} = the residual lay tension
Δp_i = the difference in internal pressure compared to as-laid
A_i = the pipe internal bore area
ϑ= Poisson's ratio
A_s = the pipe cross-sectional steel area
α = the thermal expansion coefficient for the pipe material
ΔT = the difference in temperature relative to the temperature during installation

Based on the above equation, it is evident that the key design parameters affecting the estimation of the effective axial force are the residual lay tension and the difference in internal pressure and temperature relative to installation. Hence, reducing unnecessary conservatism in those parameters is instrumental in ensuring the effective axial force is not overestimated. This can be achieved as follows:

- Maximum Design Temperature:Avoid overly conservate design margin for the maximum design temperature. The design margin should reflect the level of uncertainty based on the pipeline service and level of uncertainty in upstream condition e.g. the level of uncertainty for a multiphase oil production pipeline from an offshore well, dry gas pipeline from compressor outlet and water injection pipeline from pump outletdiffers significantly and hence the design margin should be specifically evaluated for each case.

- Installation Temperature:Pipeline temperature during installation depends on the ambient seawater temperature. The installation temperature is typically conservatively considered to be the minimum seawater temperature i.e. assuming the pipeline will be installed on the coldest day throughout the year. This conservative assumption can be optimized by considering the seasonal seawater temperature during the planned window for pipeline offshore installation. Offshore installation is typically carried out during warmer months to avoid extensive weather downtime during the winter months.
- Residual Lay Tension:The residual lay tension is often conservatively considered to be zero. This is an overly conservative assumption particularly for pipelines with heavy concrete weight coating where the required lay tension would be substantial and hence the residual lay tension would be significant. Hence, the residual lay tension should be estimated based on pipeline installation finite element analysis and included accordingly in the calculation of the effective axial force.

Optimizing Pipeline Out-of-straightness. The route selection should be optimized to increase the radius of curvature at route bends as far as feasible. In addition, full pipeline length finite element models based on seabed profile and subsea crossing configuration, if any, should be used to ensure vertical out-of-straightness is accurately simulated. Seabed areas with high irregularities should be avoided, as far as practicable, to minimize the vertical out-of-straightness.

Pipe-soil Interaction Assessment. The peak lateral soil resistance is a key design parameter affecting the pipeline critical buckling force and hence should be defined based on detailed pipe-soil interaction study.The pipe-soil interaction study should be conducted by a specialized geotechnical consultant with solid understanding of the soil models applicable for subsea pipelines, which are fundamentally different than solid models for offshore structures. The soil parameters used in the pipe-soil interaction study should be defined based on sound laboratory testingof undisturbed soil samples acquired using shallow boreholes along the pipeline route and supplemented by in-situ testing such as piezocone penetration tests (PCPT). The location of the soil samples should be carefully selectedto cover the various soil types and conditions as per seabed soil zoning based on the geophysical survey results (M. Abdel Hakim et al [4]).

Estimation of Hydrodynamic Loads. Analytical lateral buckling susceptibility calculation as per DNV-RP-F110 [2] considers themaximum hydrodynamic loads (100 year and 1 year Return Period environmental conditions). Hence, wave induced velocity is often estimated based on maximum wave height. However, this approach is considered to be overly conservative due to the spatial and temporal variation of wave heights along the pipeline. Since subsea pipelines in shallow water are normally subjected to high wave induced velocities, they are generally not statically stable i.e. they are designed to accommodate limited lateral displacement during extreme storm conditions (i.e. 10 times the pipeline diameter, DNV-RP-F109 [3]). Hence, applying the maximum wave height along the pipeline in the lateral buckling finite element analysis (which is a static and not a dynamic finite element analysis) would result in pipeline instability. Hence, estimating the wave induced velocity based on the significant wave height is considered to be a more realistic representation of the hydrodynamic loads.

Controlled Buckling

Controlled bucklingdesign approach is based on deliberately introducing significant out-of-straightness and / or reduction in lateral resistance to trigger lateral buckling at predetermined locations along the pipeline route.The formation of multiple engineering buckles allows for distributing the thermal expansion among the buckles thus ensuring the post-buckling bending moments and longitudinal strains are within acceptable limits.In addition, engineered buckle initiators(e,g, using steel or coated concrete supports)typically produce more benign buckles compared to uncontrolled buckles on the seabed. For high temperature high pressure pipelines where buckling prevention is difficult and / or expensive to achieve, controlled buckling represents a technically feasible and cost-effective alternative solution.

Controlled Buckling Techniques

A wide variety of techniques have been used in the industry to trigger buckling. Typical examples include the following:

- Vertical upset
- Snake lay
- Local weight reduction
- Zero radius bends
- Pre-bent sections
- Intermittent rock dumping

Controlled Buckling Design Challenges

Controlled buckling is more complex design approach compared to buckling prevention with various design challenges as detailed below.

Design Reliability. Controlled buckling relies on triggering a sufficient number of buckles to distribute the thermal expansion between the buckle sites. As the number of required buckles increases, the spacing between them decreases and hence the likelihood of triggering all planned bucklesalso reduces. This means that the robustness of the solution reduces, and the distance between actual buckles willbe longer than designed for. The as-laidpipe out-of-straightness and axial / lateral soil resistance are key parameters affecting the critical buckling force for the engineered buckles as well as unplanned buckles in-between. Hence, the uncertainty in these parameters represent a key design challenge for achieving the required reliability in buckle formation at the planned locations and / or preventing unplanned buckles between them.

Pipeline Stability at Buckle Triggers. Local reduction of pipe submerged weight is often used to increase the reliability of buckle formation and / or reduce the post buckling loads. In shallow water developments, this can lead to pipeline instability at the buckle trigger locations. It can also pose a challenge for curved laying stability at buckle triggers located on route curves.

Strain Capacity and Weld Flaw Acceptance Criteria. High post-buckling bending moments and longitudinal strains can pose serious threats to the structural integrity of the pipeline.Particular designchallenges include:

- Local buckling limit state for pipelines with high D/t ratio($D/t > 45$) where the pipe moment capacity is significantly reduced. This is typically the case for large diameter low pressure pipeline systems such as export trunklines.
- Crack growth rate in sour environment can be substantially higher than in-air environment (up to 40 times or even higher [2]). Hence, this would typically lead to strict limitson acceptable girth weld flaw sizes for sour service pipelines,which would adversely affect the pipelay campaign due to high weld repair rate. Consequently, the maximum tensile strain in post-buckling configuration is maintained at considerably lower limit compared to other limit states such as local buckling or plastic collapse.

Pipeline Walking and Buckling Interaction. Lateral buckles effectively divide a 'long' pipeline into a series of 'short' pipeline sectionssince the pipeline no longer reaches full axial restraint along these sections between the lateral buckles. Hence, these short pipeline sections can be susceptible to pipeline walking due to temperature transient during start-up.The likelihood of walking is particularly high for the section between the pipeline hot end and the first buckle since this section experiences the steepest temperature gradient.

The pipeline walking and buckling interaction can lead to overstressingof the riser tie-in spool due to the accumulated axial displacementover recurring shutdown / start-up cycles and / or excessive buckle growth due to increase in axial feed-in to the buckles.

Delay to Pipeline Design, Procurement and Construction. The controlled buckling design approach can typically require changes to the pipeline design including pipe wall thickness, concrete weight coating thickness / density, route modifications etc.Such changes affect other design activities such as on-bottom stability, free span analysis, crossing design, material requisitions etc.Given the intensive finite element analysis requiredfor controlled buckling design, such changes areoften made at an advanced stage during design.This can be further compounded in case of late availability of bathymetric data and pipe-soil interaction design parameters.Hence, this can lead to substantial delays to the pipeline detailed design, procurement of pipeline material e.g. line pipe and sacrificial anodes as well as pipe coating activities.

Controlled Buckling Success Factors

The feasibility and reliability of controlled buckling as a design approach can be significantly improved through adopting the following best engineering practices:

Reducing Lateral Resistance at Buckle Triggers. Utilization of engineered buckle triggers such as concrete or steel supports can improve the design reliability compared to engineered buckles on seabed using snake lay since the pipe-support interface friction is known with significantly higher certainty compared to soil resistance. Pipe-support interface friction is based on simple coulomb friction whilesoil resistance comprises of a frictional component and a passive resistance component based on pipe embedment which is affected by multiple factors including soil properties, pipe properties, loading history during pipeline installation etc.In addition, lateral resistance can be reduced by local reduction of the pipe weight (e.g. by reducing the concrete weight coating) and / or by reducing the pipe-support interface friction by utilizing special coating for the supports. The reduction of lateral resistance at the engineered buckles improves the design reliability by reducing the critical buckling force as well asincrease the tolerable virtual anchor spacing by reducing the post-buckle bending moment and longitudinal strain.

Reducing Feed-in to Buckles. Reducing the feed-in to the buckles can reduce the number of required engineered bucklesby reducing the post-buckle bending moment and longitudinal strain.This can be achieved by increasing the pipe weight (e.g. by increasing the concrete weight coating) between the buckle triggers to increase the axial resistance. The additional axial restraint is also beneficial for controlling pipeline walking rates, This is particularly useful for the pipeline section between the hot end and the first buckle trigger to control pipeline walking.

Increasing Strain Capacity. The tolerable virtual anchor spacing for buckle triggers can be improved by locally reducing the concrete weight coating thickness to reduce strain intensification due to stiffness discontinuity at field jointsand/or increasing the pipe wall thickness at the buckle trigger locations.In addition, the strain concentration should be assessed utilizing finite element analysis to derive strain concentration factors. Alternatively, strain concentration factor can be analytically derived based on elastic stress concentration factor as per DNV-RP-F105 [5] and Neuber's method as per DNV-RP-F108 [6].

Dynamic On-bottom Stability Analysis. The local reduction of concrete weight coating at buckle triggers typically means that the pipeline would not meet the virtual stability criteria and hence would require additional stabilization on both sides of the buckle triggers. The requirement for additional stabilization can be significantly reduced or even eliminatedby utilizing dynamic on-bottom stabilitytoquantify thepipe lateral displacementand designing the support length to accommodate the pipeline displacement.

Utilizing Subsea Crossings as Buckle Triggers. Subsea crossings introduce vertical out-of-straightness in the pipeline and hence increase the pipeline susceptibility to lateral buckling. This would represent a design challenge for buckling prevention design approach. On the other hand, subsea crossings can be used as engineering buckle triggers and thus represent an integral part of the controlled buckling scheme.

Acceleration of Lateral Buckling Design. The lateral buckling design can be accelerated by utilizing short finite element models i.e. virtual anchor spacing (VAS) models to determine the critical buckling force and post buckle response for various buckle triggers as well as rogue buckles on the seabed. These models require substantially less effort and time compared to full length models and thus should be used to develop the controlled buckling scheme. Full length finite element models should only be used at the end to verify the design as well as assess pipeline walking and buckling interaction. In addition,earlystart of geophysical and geotechnical surveys is also highly important to ensure timely availability of bathymetric survey and pipe-soil interaction information to avoid delays to the lateral buckling design.

Conclusions

Buckling prevention can provide a simple and inherently safer solution for pipelines characterized by modest compressive axial forces(relatively low operating temperature and / or small diameter) and with limited out-of-straightness (relatively flat seabed and no / limited crossings).On the other hand, controlled buckling is an industry proven and cost-effective solution for pipelines characterized by high compressive axial forces (high operating temperature and / or large diameter) and pipelines with significant out-of-straightness (due to route curvature, multiple subsea crossings,seabed irregularity,etc.).Buckling prevention can also be used in combination with controlled buckling for long pipelines towards the cold end of the pipeline.

The key merits and shortcomings of both design approaches are summarized as follows:

Buckling Prevention Merits

- Inherently safer design compared to controlled buckling due to elimination of lateral buckling and associated integrity risks.
- Simple design solution which also avoids additional offshore construction activities such as installation of buckle triggers.
- The increase in pipe submerged weight to suppress buckling helps in suppressing pipeline walking as well.
- Avoid design and construction challenges associated with high strain levels typically associated with controlled buckling.

Buckling Prevention Shortcomings

- A significant increase of concrete weight coating thickness is typically required to ensure design reliability and avoid risk of uncontrolled buckling, which may exceed practical limits for high temperature pipelines.
- Increase in free span rectification costs due to reduction in allowable span length as a result ofthe increase in pipeline submerged weight.
- Limited flexibility to tolerate any future escalation in operating temperature.

Controlled Buckling Merits

- Proven and cost-effective solution for pipelines with significant out-of-straightness due to field congestion, seabed irregularity and / or multiple crossings.
- Concrete or steel sleepers are simple to fabricate and can be tailored as per design requirement.
- Offshore installation of sleepers is a straightforward operation for installation contractors.
- Subsea crossings can be utilized as engineered buckle triggers to optimize mitigation costs.

Controlled Buckling Shortcomings

- Post buckle strains and pipeline displacement can pose serious threat to pipeline integrity if not carefully assessed and mitigated.
- Reduced allowable weld flaw sizes due to high tensile strains and cyclic stresses.
- Requirement for large crossing supports to accommodate pipeline lateral displacement at subsea crossings used as buckle triggers.
- Requirement for additional stabilization to compensate for reduced concrete weight coatingthickness at subsea crossings used as buckle triggers.
- Requirement to mitigate pipeline walking instigated due to buckling.

The above conclusions are presented with the aim of providing guidance rather than prescribe off-the-shelf solutions. The designer should establish the optimum design solution on a case-by-case basis based on detailed techno-economic evaluation in order to answer the fundamental question of whether to aim for buckling prevention, controlled buckling or a combination of both solutions.

References

1. Hobbs, R.E. 1984, In-service Buckling of Heated Pipelines, *ASCE Journal of Transportation Engineering* Vol. **110** No. 2.
2. Det Norske Veritas Recommended Practice DNV-RP-F110, 2020, Global Buckling of Submarine Pipelines.
3. Det Norske Veritas Recommended Practice DNV-RP-F109, 2021, On-bottom stability design of submarine pipelines, cables and umbilicals.
4. Abdel Hakim, M. et al, 2018, Cost Savings for Subsea Pipelines Design Enhanced Pipe-Soil Interaction Assessment, *ADIPEC*.
5. Det Norske Veritas Recommended Practice DNV-RP-F105, 2021, Free spanning pipelines.
6. Det Norske Veritas Recommended Practice DNV-RP-F108, 2021, Assessment of flaws in pipeline and riser girth welds.